

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	10 July 2026
Team ID	SWTID-2026-6119
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Functional Requirements:

These are the functional capabilities the solution needs to deliver.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Applicant Data Input	Present a web form that collects applicant demographic and financial details, rejecting submissions that leave a required field empty.
FR-2	Credit Approval Prediction	Run the submitted values through the saved encoder/scaler before passing them to the trained model, which returns an Approved or Rejected call.
FR-3	Result Display	Render the decision on a results page and surface a short explanation of the inputs that weighed most on the outcome.
FR-4	Model Training Pipeline	Pull in and clean the historical approval dataset, then fit and score several candidate models — Logistic Regression, Decision Tree, Random Forest and XGBoost.
FR-5	Model Comparison & Selection	Rank the candidate models on Accuracy, Precision, Recall and F1-Score, and save the top performer to disk as model.pkl.
FR-6	Error Handling	Catch missing or malformed form submissions without crashing, and route unknown URLs to a custom 404 page.

Non-functional Requirements:

These are the quality attributes the solution is expected to meet.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	A bank officer with no technical background should be able to enter applicant details and read the outcome without any training session.
NFR-2	Security	Financial details entered on the form must not be logged or kept around any longer than the single prediction that needs them.
NFR-3	Reliability	Well-formed input should always yield an Approved/Rejected result, and the system should degrade gracefully when a field is missing or malformed.
NFR-4	Performance	A prediction should come back within roughly 2-3 seconds of submission when run on typical hardware.
NFR-5	Availability	The Flask app should stay up throughout normal testing and demo sessions, with only brief, infrequent downtime.
NFR-6	Scalability	The preprocessing-plus-inference pipeline should be written so new applicant features can be folded in later without a rebuild.